Live Bird Price Prediction Dashboard

Model Evaluation Results

Evaluation setup

The project README reports a time-aware evaluation using the last 90 days as the test window (2026-08-03 to 2026-10-31). Backtesting used rolling-origin 7-day forecasts every 3 days across districts, with 980 forecast points reported. The Random Forest uses 300 trees and produces a recursive 7-day forecast.

Data limitation: The repository states that all data is synthetic. These results demonstrate the method on generated data and must not be presented as validated real-market forecasting performance.

1. Overall model comparison

Model	MAE (■/kg)	RMSE (■/kg)	MAPE (%)
Naive (last price carried forward)	1.71	2.28	2.09
Seasonal naive (same weekday last week)	2.06	2.69	2.52
Random Forest	1.64	2.15	2.02

Metric interpretation

MAE (Mean Absolute Error): average absolute difference between predicted and actual prices, in ■/kg. **RMSE (Root Mean Squared Error):** error metric that gives relatively more weight to larger errors. **MAPE (Mean Absolute Percentage Error):** average absolute percentage error.

On the reported synthetic-data test, Random Forest has the lowest overall MAE (■1.64/kg), RMSE (■2.15/kg), and MAPE (2.02%). Its MAE is only ■0.07/kg lower than the naive baseline (■1.71/kg), so the improvement is marginal. The README notes that gains were not tested for statistical significance.

2. MAE by forecast horizon

Horizon	Naive MAE (■/kg)	Random Forest MAE (■/kg)
Day 1	1.10	0.98
Day 2	1.37	1.31
Day 3	1.70	1.66
Day 4	1.66	1.74
Day 5	1.96	1.88
Day 6	2.12	1.99
Day 7	2.02	1.96

Random Forest has lower MAE than the naive baseline on Days 1, 2, 3, 5, 6 and 7, but performs worse on Day 4. This supports the README's conclusion that the model is only marginally better overall on the synthetic dataset.

3. Conclusion and limitations

The reported backtest suggests the Random Forest can learn some patterns in the generated data, but its gain over the simple last-price baseline is small. The results should be treated as a prototype evaluation, not proof of real

poultry-market accuracy. Before real deployment, evaluate on verified market-price data across more seasons and districts, include relevant external drivers, and review forecast intervals and harvest rules.

Source: project README.md, Model Evaluation section. Values in this report reproduce the metrics documented in the README; this report does not independently rerun the training pipeline.